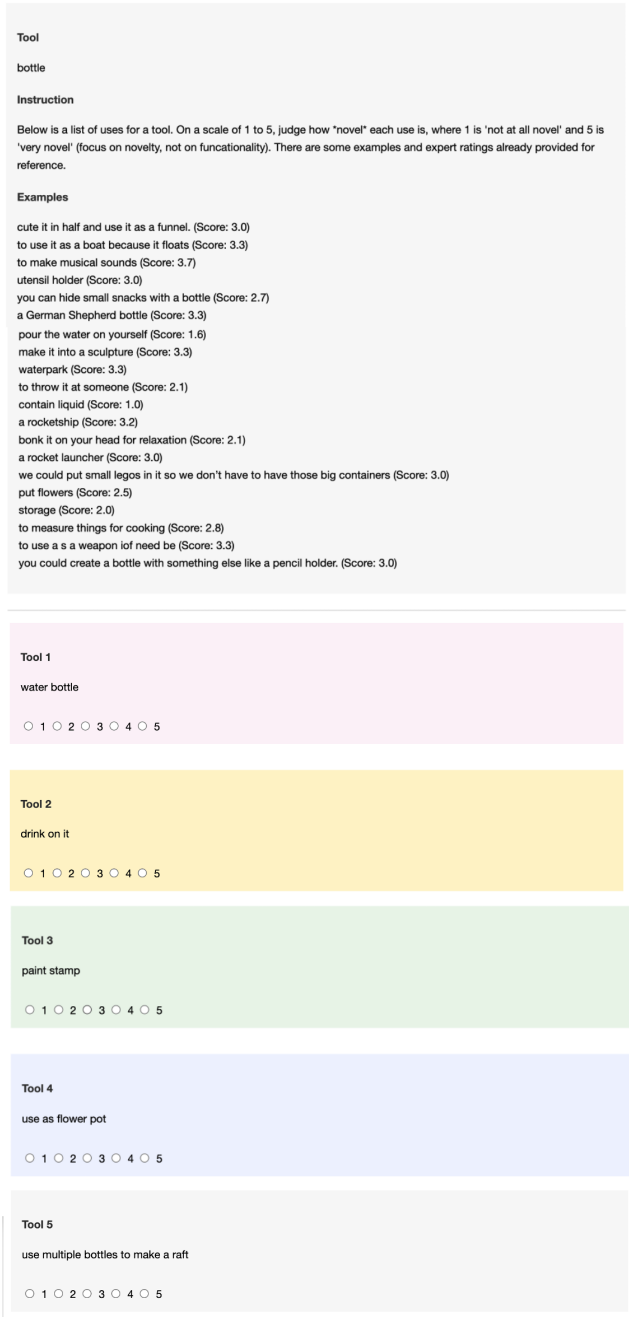

Figure 5: Annotation interface for AUT.

Question

What would happen if plastic could biodegrade as quickly as paper?Let's think step by step.

Answer

Sure, let's think through this step by step:

1. **Current Context:** Currently, plastic takes hundreds of years to biodegrade, while paper biodegrades much more quickly, usually within a few months.

2. **Assumption**: If plastic could biodegrade as quickly as paper, it would significantly reduce the time it takes for plastic waste to break down.

3. **Environmental Impact**:

   - **Soil**:
     - Faster breakdown of plastic would mean less contamination of soil over time.
     - Plants grown in such soil might suffer less from plastic particles blocking water and nutrient absorption.
   - **Landfills**:
     - Landfills would be less burdened by plastic waste, as it would decompose more rapidly.
     - This would reduce the volume of waste in landfills and the associated environmental concerns of waste seepage and methane production.
   - **Microplastics**:
     - The biodegradation of plastic could reduce the accumulation of microplastics in the environment.
     - Microplastics are harmful to marine and land wildlife, and their reduction would significantly improve ecosystem health.

4. **Economic Impact**:

   - **Waste Management**:
     - The cost of waste management and cleanup would likely decrease, as there would be less need for long-term storage and disposal strategies.
     - This could lead to reduced costs for governments as well as individuals and businesses.
   - **Recycling Industry**:
     - The reduced volume of post-consumer plastic waste might impact the recycling industry, potentially making it less necessary or economically viable to recycle as much plastic.
     - However, it could also create a new market for companies that specialize in breaking down and reusing degraded plastic.

5. **Technological and Societal Responses**:

   - **Innovation**:
     - There might be less urgency to develop alternative materials to plastic if traditional plastic could easily biodegrade.
     - However, this could also spur innovation in sustainable and biodegradable materials to ensure we are still environmentally responsible.
   - **Behavioral Changes**:
     - People might still use plastics more liberally due to the reduced long-term implications, necessitating continued public education on the importance of proper disposal and reduction.
   - **Regulations**:
     - Governments might ease plastic usage regulations if plastic biodegradation is seen as a sufficient solution to environmental concerns.
     - Regulations could shift toward encouraging

Rubric

- Definition: This task focuses on the ability to foresee consequences or outcomes of an unusual or hypothetical situation. For example, what would be the implications if animals could talk?
- Example: What would happen if the world's deserts suddenly turned into forests?Let's think step by step.

Dimension	What to Look For	1	3	5
Fluency	Number of consequences	1–2	6–8	≥12 distinct effects
Flexibility	Diversity of domains (social, ecological, economic, legal, technological, moral …)	1–2 domains	3–4	≥6 domains
Originality	Novelty of consequences	generic ("laws against noise")	some less-common links	multiple specific and unexpected but plausible chains ("AI translation obsolete because parrots become spies")
Elaboration	Causal depth and mechanism	statement only	"because→therefore" chain or example	layered reasoning with second-order effects or quantitative hint

Fluency

○ 1 ○ 2 ○ 3 ○ 4 ○ 5

Flexibility

○ 1 ○ 2 ○ 3 ○ 4 ○ 5

Originality

○ 1 ○ 2 ○ 3 ○ 4 ○ 5

Elaboration

○ 1 ○ 2 ○ 3 ○ 4 ○ 5

Figure 6: Annotation interface for TTCT.

Solution

```
def solve():
    _t = int(input())
    for _ in range(_t):
        n = int(input())
        a = list(map(int, input().split()))
        count = 0
        i = 0
        while i < n-1:
            if a[i] > a[i+1]:
                count += 1
                a[i+1] += a[i]
                del a[i]
                n -= 1
                i -= 1
            i += 1
        print(count)
```

☐ if statement ☐ for loop ☐ while loop ☐ break statement ☐ continue statement ☐ pass statement ☐ match statement

☐ stack ☐ queue ☐ tuple ☐ set ☐ dictionary ☐ linked list ☐ tree ☐ graph ☐ heap ☐ hashmap

☐ two pointers ☐ sliding window ☐ matrix operation ☐ depth first search ☐ width first search ☐ back tracking

☐ divide & conquer ☐ Kadanes algorithm ☐ binary search ☐ recursion ☐ dynamic programming ☐ greedy algorithm

☐ misc ☐ minimax ☐ topological sort ☐ sorting ☐ graph traversal

Figure 7: Annotation interface for NeoCoder.